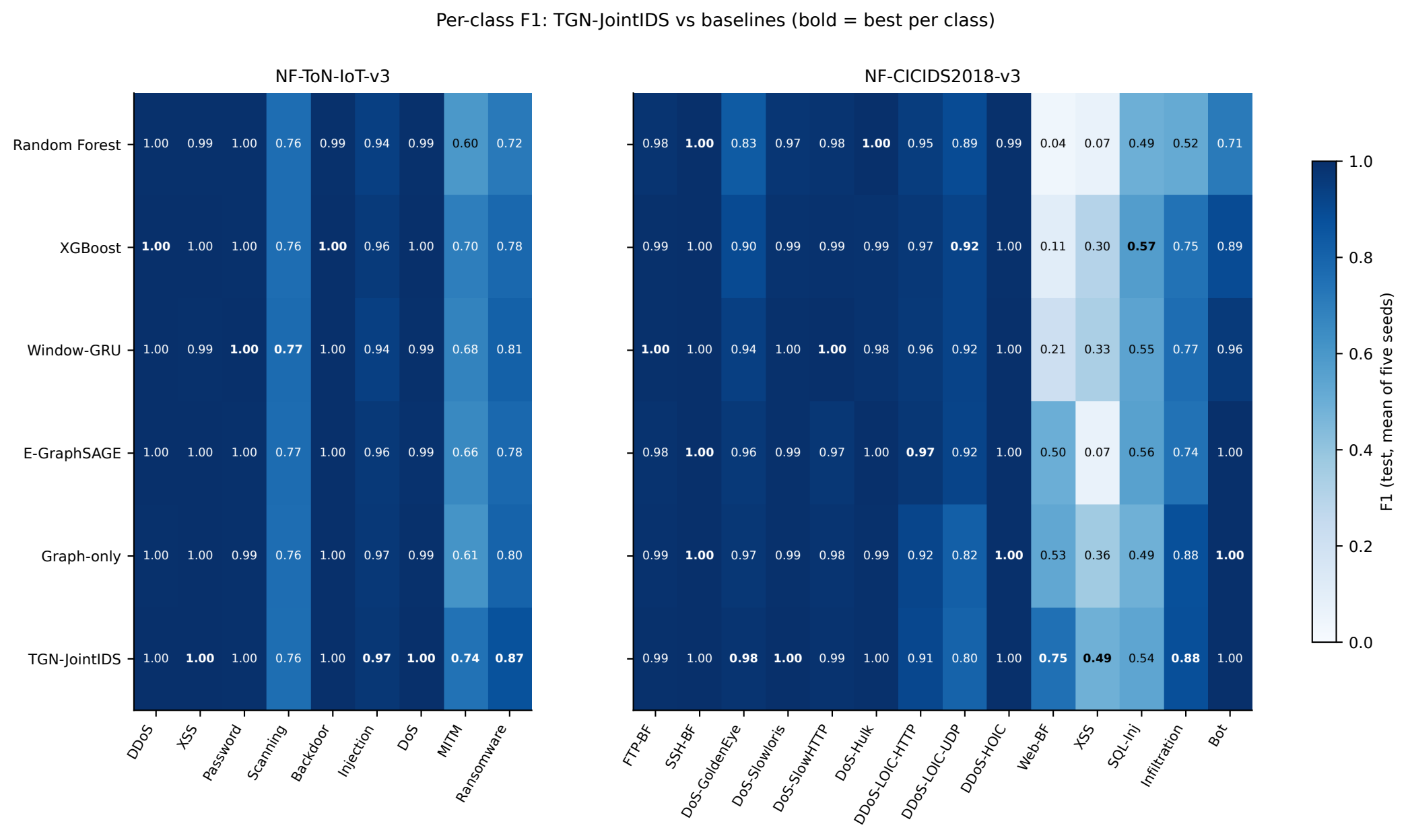
Per-class F1: TGN-JointIDS vs baselines (bold = best per class)
NF-ToN-IoT-v3
NF-CICIDS2018-v3
Random Forest
XGBoost
Window-GRU
E-GraphSAGE
Graph-only
TGN-JointIDS
1.00 0.99 1.00 0.76 0.99 0.94 0.99 0.60 0.72
1.00 1.00 1.00 0.76 1.00 0.96 1.00 0.70 0.78
1.00 0.99 1.00 0.77 1.00 0.94 0.99 0.68 0.81
1.00 1.00 1.00 0.77 1.00 0.96 0.99 0.66 0.78
1.00 1.00 0.99 0.76 1.00 0.97 0.99 0.61 0.80
1.00 1.00 1.00 0.76 1.00 0.97 1.00 0.74 0.87
0.98 1.00 0.83 0.97 0.98 1.00 0.95 0.89 0.99 0.04 0.07 0.49 0.52 0.71
0.99 1.00 0.90 0.99 0.99 0.99 0.97 0.92 1.00 0.11 0.30 0.57 0.75 0.89
1.00 1.00 0.94 1.00 1.00 0.98 0.96 0.92 1.00 0.21 0.33 0.55 0.77 0.96
0.98 1.00 0.96 0.99 0.97 1.00 0.97 0.92 1.00 0.50 0.07 0.56 0.74 1.00
0.99 1.00 0.97 0.99 0.98 0.99 0.92 0.82 1.00 0.53 0.36 0.49 0.88 1.00
0.99 1.00 0.98 1.00 0.99 1.00 0.91 0.80 1.00 0.75 0.49 0.54 0.88 1.00
DDoS
XSS
Password
Scanning
Backdoor
Injection
DoS
MITM
Ransomware
FTP-BF
SSH-BF
DoS-GoldenEye
DoS-Slowloris
DoS-SlowHTTP
DoS-Hulk
DDoS-LOIC-HTTP
DDoS-LOIC-UDP
DDoS-HOIC
Web-BF
XSS
SQL-Inj
Infiltration
Bot
F1 (test, mean of five seeds)
1.0
0.8
0.6
0.4
0.2
0.0